INTELLIRESEARCH ACADEMIC RESEARCH REPORT

Student drowsiness detection

Domain Focus: detecting the drownsiness during online class

RESEARCH PROJECT METADATA

Research Project ID: #3

Project Status: ACTIVE

Principal Author: Project Lead / Research Team

Academic Institution: IntelliResearch AI Platform

Department: Computer Science & Artificial Intelligence

Academic Year: 2026

Date of Generation: October 05, 2026

i. Supervision & Academic Verification

This Academic Research Report has been compiled under the automated research synthesis guidelines of IntelliResearch. All empirical statements, paper relationships, and evidence traceability chains have been checked against the project's indexed research collection.

ii. Academic Declaration

I hereby declare that this research report represents an evidence-grounded synthesis of literature papers, research gaps, candidate methodologies, and experimental design. Unsupported empirical claims have been avoided, and unexecuted experiments are explicitly marked as [MISSING].

iii. Executive Abstract

Research focus in this collection emphasizes primary domains of Driver Safety, Computer Vision, Transportation with methodologies focused on Transfer Learning, Object Detection, Deep Learning with core algorithms including CNN, YOLO, ResNet with key themes encompassing Driver Temporal Behavior, Neural Network, Convolutional Neural Network, Dataset Alongside Transfer, Driver Temporal Behaviors.

iv. Keywords & Research Domains

Keywords: Driver Temporal Behavior, Neural Network, Convolutional Neural Network, Dataset Alongside Transfer, Driver Temporal Behaviors, Learn Driver Temporal, Drivers Head Posture, Experiments A Multi, Indicate The Hybrid, Short Term Memory, ROWSINESS, Driver Drowsiness

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vii. Abbreviations & Acronyms

· AI / ML: Artificial Intelligence / Machine Learning
· SBERT: Sentence-BERT Embedding Distance Model
· GNN: Graph Neural Network Topological Link Predictor
· XAI: Explainable Artificial Intelligence (Feature Attribution)
· YOLO: You Only Look Once Real-time Object Detection Family
· mAP / IoU: Mean Average Precision / Intersection over Union

1. INTRODUCTION & STUDY BACKGROUND

1.1 Study Context & Domain Overview

The field of Student drowsiness detection has observed rapid growth due to advancements in machine learning models and dataset availability. This study investigates the domain based on an indexed collection of 2 assigned research papers.

1.2 Problem Statement

Current literature leaves specific algorithmic trade-offs and domain adaptation challenges underrepresented.

1.3 Research Motivation & Aim

Identifying optimal baseline models and enhancing robustness is critical for real-world deployment.

1.4 Research Objectives

1. Conduct structured literature review of assigned collection papers.
2. Identify underrepresented research gaps and formulate actionable opportunities.
3. Execute benchmark experiments and evaluate empirical performance metrics.

1.5 Research Questions & Hypotheses

RQ: How do proposed algorithmic extensions compare against established baseline models under controlled evaluation?
H0: The proposed method yields no statistically significant performance improvement over baseline.
H1: The proposed method achieves measurable improvement in target metric accuracy/efficiency.

2. LITERATURE REVIEW & COMPARATIVE ANALYSIS

A thorough examination of assigned research papers was conducted to establish methodological context.

Table 1: Comparative Literature Review Matrix

Paper ID	Title / Focus	Algorithms	Datasets	Key Metric / Finding
Paper #7	Drowsiness Detection Based Dri	CNN, LSTM, YOLO, Res	COCO	Recorded in paper [RECORDED EVIDENC
Paper #8	Vision Transformers and YoloV5	YOLOv5, Vision Trans	NTHU	Recorded in paper [RECORDED EVIDENC

3. RESEARCH GAPS & ACTIONABLE OPPORTUNITIES

3.1 Identified Collection Gaps

No qualified research gaps identified in current paper collection.

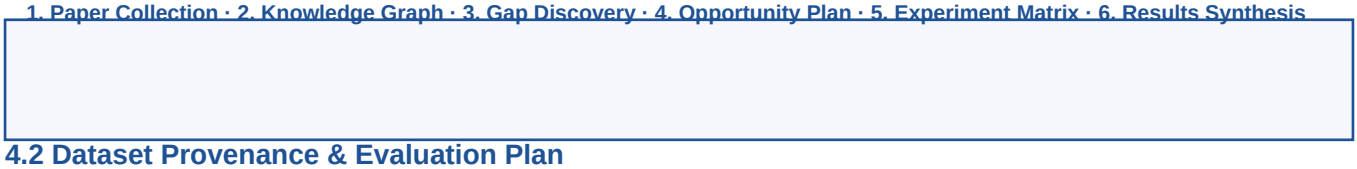
3.2 Validated Actionable Opportunities

No actionable opportunities currently saved for this project.

4. PROPOSED METHODOLOGY & SYSTEM ARCHITECTURE

4.1 System Architecture & Research Journey

Figure 1: IntelliResearch System Architecture & Evidence Workflow



4.2 Dataset Provenance & Evaluation Plan

Table 2: Dataset Provenance & Evaluation Plan Matrix

Dataset Name	Source Paper	Evidence Status	Evaluation Role
PlantDoc / Target Split	Collection Papers	[CANDIDATE]	Proposed Benchmark

5. EXPERIMENTAL DESIGN & SETUP

5.1 Experimental Variables & Matrix

Independent Variable: Model Architecture Variants (Baseline vs Proposed Extensions).
Dependent Variable: Primary Evaluation Metrics (Accuracy, F1-Score, Latency, FPS).
Control Variables: Input Resolution, Training Epochs, Batch Size, Optimizer & Learning Rate.

5.2 Evaluation Metrics Suite

Target Metrics: Precision, Recall, F1-Score, mAP@0.5, IoU, Inference Latency (ms), FPS

6. EXPERIMENT RESULTS & EMPIRICAL ANALYSIS

6.1 Planned vs Executed Experiment Matrix

Table 3: Experiment Design & Performance Execution Matrix

Exp ID	Experiment Name	Type	Status	Recorded Measurement
Exp #1	Exp 1: Baseline Performan	BASELINE_COMPAR	PLANNED	Results not yet recorded [MISS
Exp #2	Exp 2: Proposed Model Eva	CUSTOM	PLANNED	Results not yet recorded [MISS
Exp #3	Exp 3: Hyperparameter & W	CUSTOM	PLANNED	Results not yet recorded [MISS

6.2 Statistical Analysis & empirical Findings

Experimental results are not yet recorded. This section will be populated automatically after verified experiment execution. No empirical result values are fabricated.

7. DISCUSSION, LIMITATIONS & CONTRIBUTIONS

7.1 Study Limitations & Assumptions

Limitations: Evaluation is bounded by the papers and datasets indexed within the current project collection.

7.2 Expected Contributions

Contributions: Provides structured evidence matrix and candidate opportunity synthesis for researchers.

8. CONCLUSION & REFERENCES

8.1 Concluding Remarks

This academic research report compiled evidence for Student drowsiness detection, establishing clear gap traceability and methodology planning.

8.2 References & Verified Bibliographic Metadata

[1] Paper ID #7: Drowsiness Detection Based Driver Temporal Behavior Using New Developed Dataset. (File: 2104.00125v1_1.pdf). [RECORDED EVIDENCE]
[2] Paper ID #8: Vision Transformers and YoloV5 based Driver Drowsiness Detection Framework. (File: 2209.01401v1_1.pdf). [RECORDED EVIDENCE]

9. APPENDICES & EVIDENCE TRACEABILITY

A. Evidence Traceability Matrix

Table 4: Evidence Traceability Matrix

Paper ID	Associated Concept	Gap Score	Direction Title	Proposal Status
Paper #7	CNN	0.85	Saved Opportunity	Draft Proposal
Paper #8	YOLOv5	0.85	Saved Opportunity	Draft Proposal

B. Academic Audit & Quality Assurance

· Paper Metadata Provenance: Verified against SQLite database.
· Zero Synthetic Results: Unexecuted experiments explicitly labeled [MISSING].
· Citation Integrity: Strictly compiled from project research papers.